

Rigorous Reasoning in Maths

Fact $\rightarrow$ The limit of a fraction is not limit of fractions

$\Rightarrow \sqrt{2}$ is irrational (proof)

$$\frac{p}{q}$$

$$p, q \in \mathbb{Z}$$

$$q \neq 0$$

p, q are co-prime.

$$\sqrt{2} = p/q$$

$$p^2 = 2q^2$$

has to be even

$$p = 2k$$

$$4k^2 = 2q^2$$

q has to be even as well

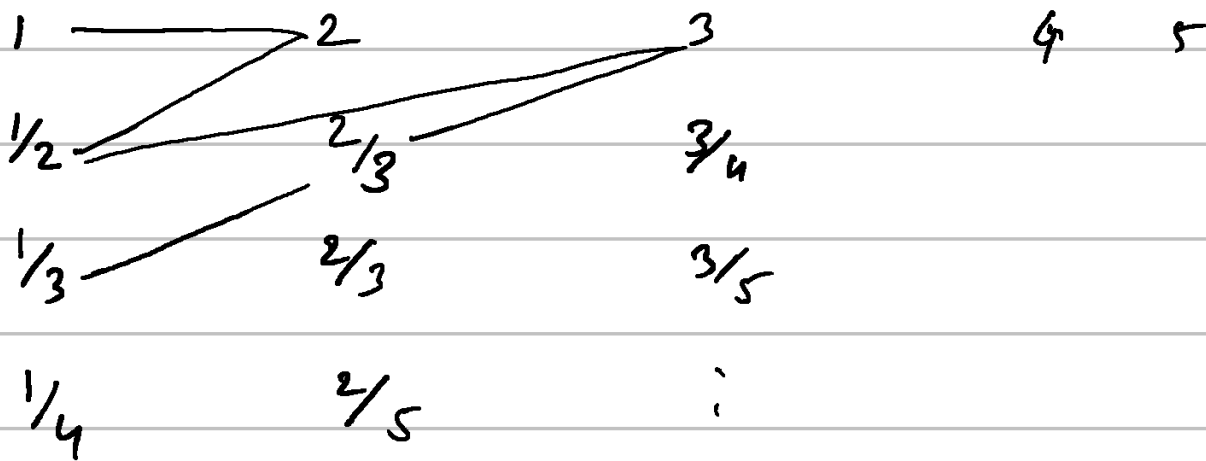
$$q = 2n$$

But if $\boxed{\begin{array}{l} p = 2k \\ q = 2n \end{array}} \rightarrow \text{not coprime!}$

Hence, our assumption that we could represent $\sqrt{2}$ as p/q .

What is true $\rightarrow$

- ① Q has as many as Z or $\checkmark$
- ② Q has more than Z .



1 — 1 mapping b/w Q & Z.

same cardinality!

Terminology

- Group $\rightarrow$ G Associative binary operation. +
(G)

$$g + g' \in G$$

$$(g + g') + g'' = g + (g' + g'')$$

② $0 \rightarrow$ identity element

$$g + 0 = g$$

③ $-a \rightarrow$ inverse exists

$$g + (-g) = 0$$

• Ring (R)

① A commutative group.

$$g + g' = g' + g$$

② Binary operation " $\times$ " exists

$$g \times g' \in G$$

③ existence of identity 1
 $g \times 1 = g$

(iv) Operation $\times$ is distributive wrt the first

$$g \times (g' + g'') = g \times g' + g \times g''$$

eg $\mathbb{Z}$ is a ring.

If there are ^{same} 2 $\uparrow$ identity elements then ring is $\{0\}$.
 $k \rightarrow$ identity

$$x \times k = x$$

$$x + k = x$$

$$(x + k) \times x = x \times x$$

$$x \times x + k \times x = x \times x$$

$$x \times k = 0 = x$$

So

$$\boxed{x = 0}$$

• Field (K)

① A ring with $0 \neq 1$

② A multiplicative inverse exists " $\frac{1}{x}$ " except 0

$$g \cdot \frac{1}{g} = 1$$

Prove that $\{\emptyset, \{\ast\}\}$ is a field
 $\downarrow$
 null set

with ops Δ (symmetric difference) &
 $\otimes$ (XOR)

& $\cap$ & identity element $\emptyset, \{\ast\}$

$$\begin{aligned} 0 &\rightarrow \emptyset \\ 1 &\rightarrow \{\ast\} \end{aligned}$$

$$\begin{aligned} \emptyset \Delta \emptyset &= \emptyset \\ \{\ast\} \cap \emptyset &= \emptyset \end{aligned}$$

Vector Spaces

⇒ A vector space over a field $\underline{K}$ is a set $\{u, v, \dots\}$

⇒ Elements are called vectors.

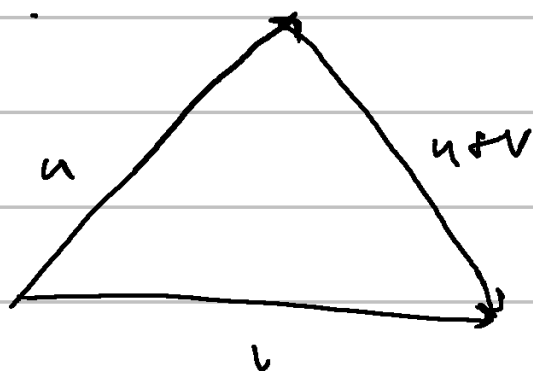
Axioms

① Binary op + Associative

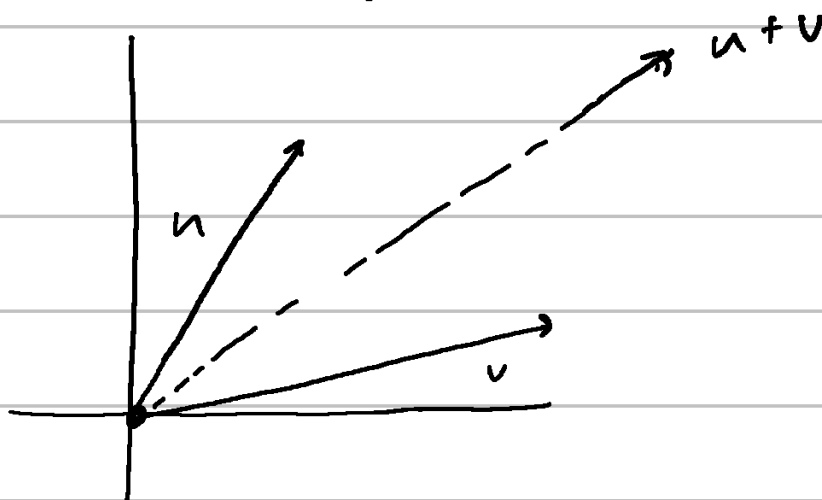
② Identity 0

③ $K \times V \rightarrow V, (\lambda, v) \rightarrow \underline{\lambda v}$
scalar multiplication.

④ $\lambda(u + v) = \lambda u + \lambda v$



not a vector
span



$\mathbb{Z}^n$ is not a vector space because
of inverses

Span $\rightarrow$ Set of vectors which can
be represented

$$v = \sum c_i v_i \quad \text{where } v_i \in V$$

$$\downarrow$$
 linear combination

$\Rightarrow$ A set of vectors is called independent if $\sum \alpha_i v_i = 0$
iff $\alpha_i = 0 \quad \forall i$

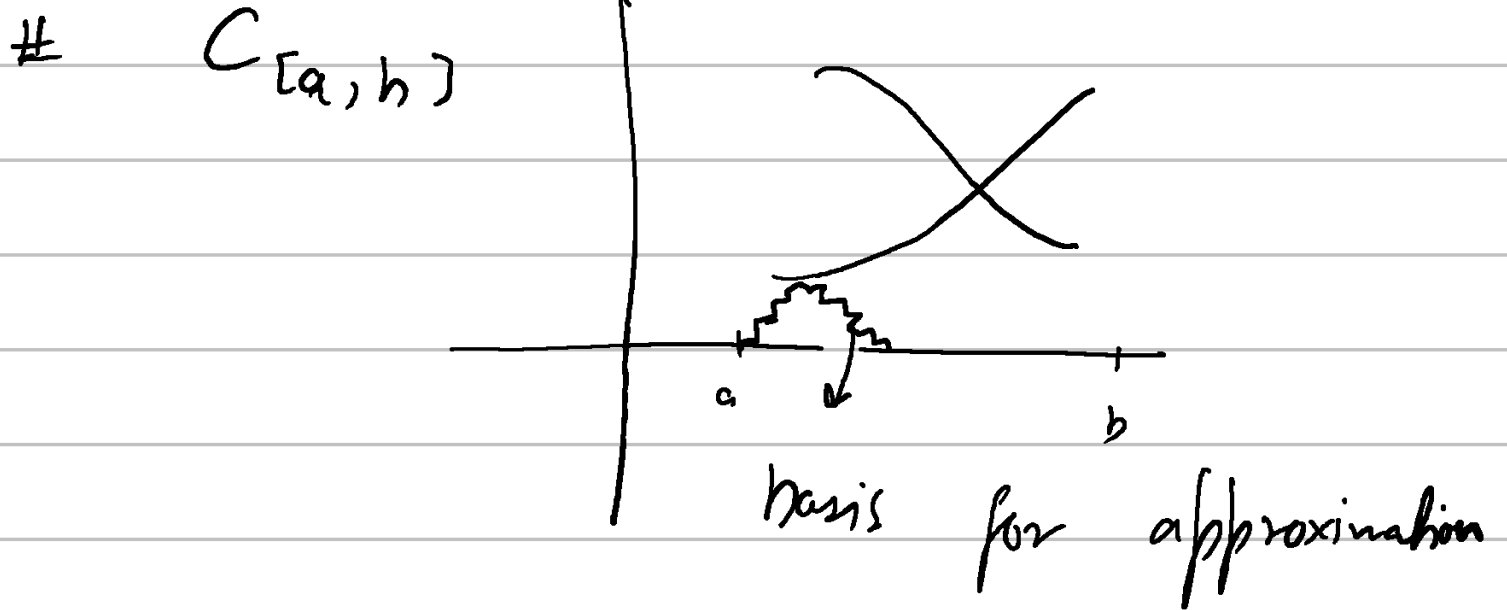
$\Rightarrow$ A n independent span $\rightarrow$ basis
of
vector space

Basis

• $\mathbb{R}^n \rightarrow$ cols of $I \rightarrow n$

• $\mathbb{C}^n \rightarrow$ cols of I & cols of $i \times I$
 $\hookrightarrow 2n$

• $\mathcal{P}_n \rightarrow$ Same as $\mathbb{R}^n + 1$



$\Rightarrow$ All bases of vector space are same dimension called dimension

$\dim(V)$

$\Rightarrow$ Read Steinitz lemma

Rank of $V = \dim(\text{span}(V))$

$$\text{rank}(\{v_1, \dots, v_n\}) := \dim \text{span}(\{v_1, \dots, v_n\})$$

- hyperplane $\rightarrow$ $n-1$ dimension space in n dimensional space.

Direct Sum

$$E, E' \subseteq V$$

$$V = E \oplus E' \quad \text{if} \quad \begin{aligned} &V = E + E' \quad \&\& \\ &E \cap E' = \{0\} \end{aligned}$$

$$\Rightarrow V = E \oplus E'$$

$e \quad e'$

$$v = \lambda e + \lambda' e'$$

Let's say $\mu \varepsilon + \mu' \varepsilon' = V$

$$\underbrace{(\lambda e - \mu \varepsilon)}_{\downarrow E} + \underbrace{(\lambda' e' - \mu' \varepsilon')}_{\downarrow E'} = 0$$

So, they have to be zero
since that is their only interaction

$$\begin{aligned} \lambda e &= \mu \varepsilon \\ \lambda' e' &= \mu' \varepsilon' \end{aligned}$$

Projections

$$\left\{ \begin{array}{l} V \rightarrow E \\ v \mapsto \lambda e \\ \downarrow \\ \text{unique} \end{array} \right. \quad \otimes \quad \left\{ \begin{array}{l} V \rightarrow E' \\ v \rightarrow \lambda' e' \end{array} \right.$$

Inner Products

• $\langle x, y \rangle = x^T y = \sum x_i y_i$

$\begin{matrix} & (x^T y) \\ & \downarrow \downarrow \\ \text{en français} \end{matrix}$

- ① Bilinear $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$
② Symmetric $\langle x, y \rangle = \langle y, x \rangle$
③ Positive definite $x \rightarrow \langle x, x \rangle$ is true
 $\langle x, x \rangle = 0 \Rightarrow x = 0$

• Norm $\|x\| = \sqrt{\langle x, x \rangle}$

- ① injection
② Linear $\|\lambda x\| = |\lambda| \cdot \|x\|$
③ triangle inequality $\|x + y\| \leq \|x\| + \|y\|$

proof by triangle inequality.

$\Rightarrow \int_a^b f \cdot g$ is an inner product for $C[a, b]$

$$\langle f, g \rangle = \int_a^b f \cdot g$$

$$\langle \lambda f, g \rangle = \int_a^b \lambda \cdot f \cdot g = \lambda \cdot \int_a^b f \cdot g$$

$$\langle f, g \rangle = \int_a^b f \cdot g = \int_a^b g \cdot f = \langle g, f \rangle$$

$$\langle f, f \rangle = \int_a^b f^2 \geq 0$$

$$\int_a^b f^2 = 0 \Rightarrow f = 0 \text{ in } [a, b]$$

$\hookrightarrow$ only for continuous.

⇒ Inner product for P_n

eg ① $\int f \cdot g$

$$\textcircled{2} \quad \left\langle \sum_{i=0}^n a_i x^i, \sum_{i=0}^n b_i x^i \right\rangle \\ = \sum_{i=0}^n a_i b_i$$

Cauchy - Schwartz inequality

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

equal only when linear dependence.

⇒ Read about proof

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|}$$

$$\begin{array}{ccc} \langle x, y \rangle = 0 & \rightarrow & x \text{ \& } y \text{ are} \\ \downarrow & & \text{orthogonal.} \\ \theta = \pi/2 & & \end{array}$$

$\Rightarrow$ cosine similarity $\rightarrow 1 \rightarrow$ similar
 $0 \rightarrow$ ortho.
 $-1 \rightarrow$ dissimilar

Orthonormal bases

$\Rightarrow$ Bases where all vectors have
 norm = 1

⇒ Let $\{a_1, \dots, a_n\} \rightarrow$ linearly independent vectors.

$$u_k = a_k - \sum_{j < k} \langle a_k, q_j \rangle q_j$$

$$q_k = \frac{u_k}{\|u_k\|}$$

⇒ Start with $k=1$

$$\textcircled{1} \quad u_1 = a_1$$
$$q_1 = \frac{a_1}{\|a_1\|}$$

$$\textcircled{2} \quad u_2 = a_2 - \langle a_2, q_1 \rangle q_1$$

$$q_2 = \frac{a_2 - \langle a_2, q_1 \rangle q_1}{\|a_2 - \langle a_2, q_1 \rangle q_1\|}$$

⋮

Revision

* $\mathbb{R}^n \rightarrow$ a valid vector space over $\mathbb{R}$.

$\mathbb{C}^n \rightarrow$ a valid vector space over $\mathbb{C}$

$\mathcal{P}_n \rightarrow$ we can represent $\mathcal{P}_n$
as $\mathbb{R}^{n+1}$

$$\mathcal{P}_n \cong \mathbb{R}^{n+1}$$

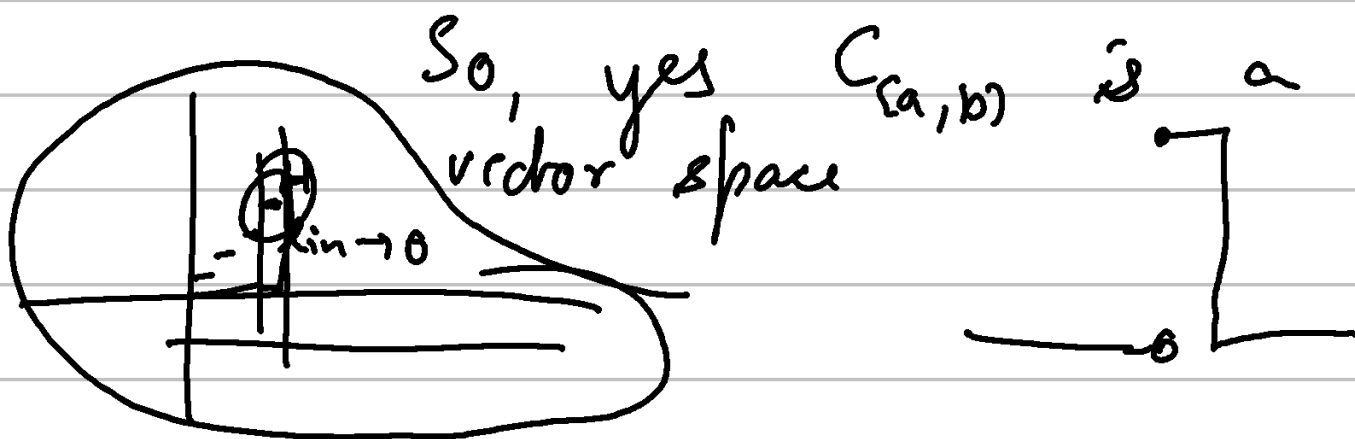
Same dimensions spaces are
always isomorphic

~~Clarify!!~~ $C[a, b] \rightarrow$ set of all continuous real functions over a closed interval $[a, b]$

$\Rightarrow$ Every continuous function can be represented as fourier series.

The functions in fourier series act as an orthogonal basis for square integrable functions

* finite area under curve.
(true for closed interval)



Steinitz exchange Lemma

→ A set of independent vectors can never have more vectors than a spanning set.

$$\{v_1, v_2, \dots, v_m\} \rightarrow \text{ind. in } V$$
$$\{w_1, \dots, w_n\} \rightarrow \text{spans } V$$

⇒ $S_0 \rightarrow \text{span } V$ ✓

$$S_j = \{v_1, v_2, \dots, v_j, w_{j+1}, \dots, w_n\}$$

Assuming a span.

$$v_{j+1} = \cancel{\sum \alpha_i v_i} + \sum \beta_i w_i$$

since
independent.

$$v_{j+1} = \sum_{i=j+1}^n \beta_i w_i$$

$$\star w_{j+1} = \frac{1}{\beta_{j+1}} \left(v_{j+1} - \sum_{i=j+2}^n \beta_i w_i \right)$$

$$\omega_{j+1} \in \text{span}(S_{j+1})$$

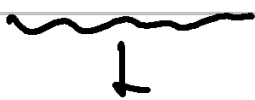
$\Rightarrow$ Since every other vector of S_j is $\checkmark$ already in S_{j+1} .

$$S_j = \{v_1, \dots, v_j, \underbrace{\omega_{j+1}}_{\in \text{span}(S_{j+1})}, \dots, \omega_n\}$$

$$S_{j+1} = \{v_1, \dots, v_{j+1}, \omega_{j+2}, \dots, \omega_n\}$$

$$S_j \subseteq \text{span}(S_{j+1})$$

$$\text{span}(S_j) \subseteq \text{span}(S_{j+1})$$



if S_j spans V then S_{j+1} also spans V .

Now, let's assume $m \geq n$

that means.

$\{v_1, \dots, v_n\}$ & there is nothing left to trade.

But $\{v_1, \dots, v_n\}$ spans V

So, $v_{n+1} \in \text{span}\{v_1, \dots, v_n\}$

But v_i are independent, so

$m > n$ is false and
 $m \leq n$ is the only case

⇒ Now using this lemma
let's try to prove that
all independent spanning
vector sets will have same
count of vectors.

⇒ $\{v_1, \dots, v_n\} \rightarrow$ independent-
(also spanning)
 $\{w_1, \dots, w_m\} \rightarrow$ spanning
(but also independent)

$$n \leq m$$

But conversely $m \leq n$

$\text{So, } m = n$

Direct Sum

$$E, E' \leq V$$

⊕ → direct sum defined as

$$E + E' = V$$

$$\& \&$$

$$E \cap E' = \{0\}$$

$$\Rightarrow E + E' = V$$

$$\forall v \in V \exists e \in E \ e' \in E' \mid v = e + e'$$

Now, to prove that the combination v, e, e' are unique. $\Rightarrow$

let's say there exist $e, e' \in E$ | $e, e' \in E'$
 $v = e_1 + e_1'$

$$e + e' = e_1 + e_1'$$

$$(e - e_1) + (e' - e_1') = 0$$

$\Rightarrow$ Since $E \cap E' = \{0\}$.

$$\begin{aligned} e &= e_1 \\ e' &= e_1' \end{aligned}$$

Projections would not exist without uniqueness

one input might have multiple outputs so not a function.

Cauchy-Schwarz Inequality

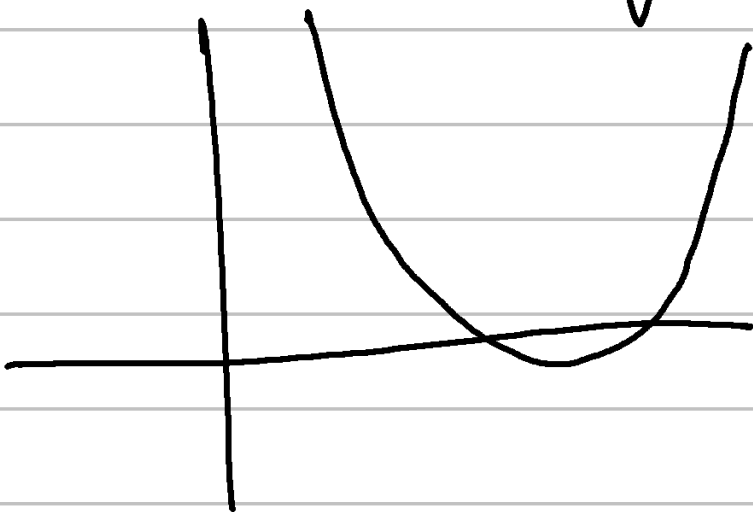
$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$$

= when linearly independent.

$$x = ty$$

$$\|x - ty\|^2 = \|x\|^2 - 2t\langle x, y \rangle + t^2\|y\|^2$$

$$\|x\|^2 - 2t\langle x, y \rangle + t^2\|y\|^2 \geq 0$$



$t \rightarrow 0$ roots $\rightarrow$ linearly independent.
 1 root $\rightarrow$ linearly dependent.
 2 roots $\rightarrow$ wtf?
 $\downarrow$
 impossible as $\|x - yt\| \geq 0$
 $\forall t \in \mathbb{R}$.

≤ 0 discriminant

$$4 \langle x, y \rangle^2 - 4 \|x\|^2 \|y\|^2 \leq 0$$

$$\boxed{\langle x, y \rangle^2 \leq \|x\|^2 \|y\|^2}$$

$$\hookrightarrow |\langle x, y \rangle|$$

